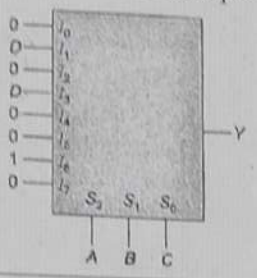
3.	Design a sequence generator for a sequence 110101. Or Design a counter for the following sequence generator $0 \rightarrow 12 \rightarrow 10 \rightarrow 6 \rightarrow 9 \rightarrow 2 \rightarrow 7 \rightarrow 14 \rightarrow 5 \rightarrow 8 \rightarrow 0$	7.5	CO3, CO4	A, E, C
4.	Design the logic circuit to perform the following operation. (a) A + B (b) B − A Or Design a sequential circuit for a MOD-10 synchronous counter. Perform the operation using state table. Draw the state diagram perform the operation using T-flip-flop.	5	CO1, CO2	U, A, R
5.	What is priority encoder? Design the 8 × 4 priority encoder with a valid output. Or Explain 8 × 1 multiplexer with the logic diagram and truth table. What is the logic function of output Y for the following 8 × 1 multiplexer?  	5	CO1, CO2	E, A, U
6.	What, in general terms, is the distinction between computer organization and computer architecture? List and briefly define the main structural components of a computer. Or Design a MOD-12 ripple-down counter and show the timing waveform with output at each clock pulse. If the input clock frequency is 600 kHz then find the frequency of the output waveform. Assume all flip flops are negative edge triggered with 0 output initially.	5	CO6	R, U
7.	What is the difference between DRAM and SRAM in terms of applications and characteristics such as speed, size, and cost? OR What do you mean by the overriding input signals? What are their types? Explain the working and operation of Jhonson counter. What are the differences between the ring and Jhonson counter?	5	CO6	R, U
8.	What are the differences among direct mapping, associative mapping, and set-associative mapping? Or Explain the operation of the SR flip-flop with logic diagram, truth table, characteristic table, excitation table, and state diagram.	5	CO6	R, U